

ISE 542

Chapter 1

The Evolution of Knowledge

Learning Outcomes

By the end of this unit the participant should be able to:

- To describe the evolution of the concept of knowledge.
- To be able to define knowledge.
- To differentiate various types and levels of knowledge.
- To explain the way in which knowledge can be converted (Nonaka's cycle of knowledge conversion).

1.1 Introduction

The new emphasis on knowledge today is significantly different from that placed on this age-old subject by Plato and Aristotle, and a host of philosophers after their time. Recently, many organizations have realized that they require more than a casual approach to corporate knowledge if they are to succeed in today's and tomorrow's economies. The perception and the reality of a new global competitiveness is a strong driving force; while rapid change and increasing competition have led organizations to seek a sustainable advantage through the managing of knowledge. They have realized that what an organization and its employees know is at the heart of how the organization functions. Knowledge itself is worthy of attention because it tells an organization how to do things and how they might do them better. As a result of this realization, they value the experience and know-how of employees, as working knowledge. Some typical examples can be drawn from industry as follows:

- A pharmaceutical company wants to cut the cycle time for the invention of new drugs and the time to market for introducing the drugs in the market. This may save millions of dollars every day. In order to achieve this, it is common to ask what knowledge resources should be provided to the scientists? In what way should the communication systems be improved? How can an environment be created that would result in increased knowledge production for the scientists?
- The top management of a company worries about its aging workforce a large proportion of which will retire over the next few years. Unless the company has a proper knowledge management plan, a significant amount of organizational experience will leave the company. Can they implement a program that will extract and transfer the knowledge of these experienced individuals to new and younger staff, before they retire?
- Staff members of a customer service centre do not share experience and know-how for handling customer enquiries with each other, resulting in hundreds of wasted hours spent resolving similar problems with little reduction in the time spent solving the customer's problems. What can be done to distribute the knowledge of the problems and solutions to other staff members?
- An experienced lawyer at a law firm possesses many years of experience pleading cases in front of several judges. When new lawyers join the law firm, no process exists to transfer knowledge about strategies that the experienced lawyer has used with each judge. Can they establish a process for transferring experience, thereby improving the firm's ability to win cases and thus enhancing its reputation?
- The management staff of the information technology

development unit realize that if they could improve the way software engineers share software coding techniques and reuse software codes they could save both time and money in software development.

The trend toward companies becoming learning organizations (LO) has also contributed to the increase in interest for knowledge. A LO is an organization skilled at creating, acquiring, and transferring knowledge and at modifying its behavior to reflect new knowledge and insights (Garvin, 1998). They have to rethink the value of knowledge once it is gone. For example:

- Experienced engineers encouraged to leave the company during downsizing or business process re-engineering periods, took valuable knowledge out of the company and had to be rehired so that essential work could continue.
- New product designers would like to replicate the success of the past, successful product designs. However, no one remembered or had recorded, what was so special about the effort the previous designers had made.
- Experienced managers proved, by their absence, to having been key knowledge brokers and assemblers. Their absence caused an enterprise a loss well beyond what their official job descriptions would suggest.
- A research and development company invests a lot of resources and money to invent what it believes to be a new technology; in fact it had been patented by its competitors a few years ago.

Many companies are now struggling to gain a better understanding of what they know, what they need to know and what to do about it. There are misconceptions that technology could replace the skill, experience and judgment of an experienced worker. Development in technology can be an enabler for knowledge management. For examples:

- The World Wide Web (WWW) has made certain forms of structured knowledge easier to collect, store in repositories, and distribute to others.
- The rise in the use of intranets is one way of expanding the role of Information Technology in communication and knowledge acquisition.
- Email provides new ways for individuals to exchange and share information and knowledge within and outside their organization.

The level of interest has been building up for several years. Many companies have long appreciated the value of knowledge to the enhance their products and customer services. The following reasons indicate why the level of interest has grown dramatically during recent years:

- **Globalization and competition** - nowadays many organizations have to rely on knowledge to create a strategic advantage for themselves. With available knowledge widely dispersed and unstructured, organizations often waste valuable time and resources in searching for the right knowledge; or they fail to access the highest quality knowledge and expertise that is available.
- **Knowledge can command a better and more secure price in the market** - Applied know-how can enhance the value of products and services. Examples are the 'smart drill' that learns how to extract more oil from an oil field, and the hotel chain that knows your personal preferences and so can give you a more personalized service.
- **Restructuring and downsizing** - Without effective mechanisms in place to capture the knowledge of experienced employees, organizations make costly mistakes or have to pay again for knowledge they once had on tap. Knowledge will be lost as the experienced employees leave the company.
- **Sharing of best practices** - If the best practices are identified and applied to similar situations elsewhere, the company can save on costs.
- **Successful Innovation** - Companies applying KM methods have found that through knowledge networking they can create new products and services faster and better.

Companies are becoming aware of the potential of technology to enhance knowledge work and aware that this can be realized only if they understand more about how knowledge is actually created and shared. Most organizations are looking for best practices, new ideas, creative synergies and breakthrough processes from making effective use of knowledge. In addition, it is clear that much of the knowledge they need already exists within their organizations but is not accessible or available when required. For example, HP CEO Lew Platt said, "if HP knew what HP knows, we would be three times as profitable", expressing a belief in the potential value of knowledge (Davenport and Prusak, 1998). Prestigious companies such as Skandia, Ernst and Young, IBM, McKinsey, Schick Asia Ltd, have employed knowledge management professionals such as a Knowledge Management Manager and Chief Knowledge Officers (CKO) to oversee the knowledge resources of their companies.

1.2 The Concept of Knowledge

The concept of knowledge has been evolving over the last two-plus millennia from justified true belief - math and logic of the Greeks, to the scholastic method of the Middle Age Era, to the much increased value

placed on it that inspired the Scientific/industrial revolution and the scientific method, during the industrial revolution. This historical perspective and understanding is critical for defining the concept of knowledge for modern organizations (Hans Siggaard Jensen, 2003). A summary is shown in Figure 1.1.

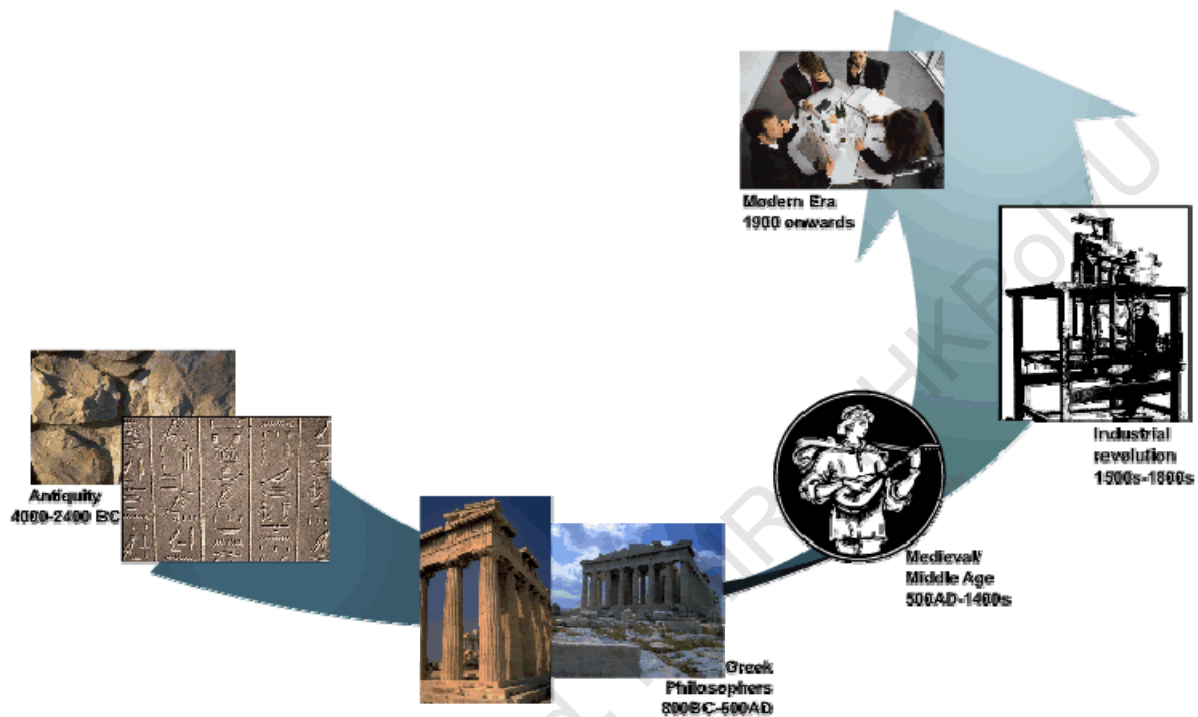


Figure 1.1 The concept of knowledge

1.2.1 Antiquity

There is little sign of knowledge concepts in antiquity. In a stone age society, people had certain practical knowledge and possibly also thoughts as to what knowledge was and what knowledge was cognizable. One can only conclude certain things on the basis that their knowledge was used to implement certain projects, etc. Myths and mythologies, large-scale narratives and symbolic depictions served the purpose of collecting and disseminating knowledge. The Babylonians and Egyptians had forms of knowledge characterized as scientific or something approximate to it (in astronomy, for example). However, much of the history of the concept of knowledge in antiquity can not be known because there are no written records. It is not until ancient Greece (5th Century BC) that more systematic forms of knowledge were developed and reflection was devoted to what knowledge actually is.

1.2.2 Greek Philosophers – Math and Logic

At that time, knowledge was defined as justified true belief or constant

knowledge. Knowledge pertains to be constant and universal; it has stability; it is something about which consensus and certainly can be achieved. Plato evolving his philosophy, and mathematicians such as Pythagoras were also devising comprehensive geometric theories and studies. Knowledge is certain knowledge and its certainly stems from all forms of logical proof. Mathematical propositions are examples of statements that can convey knowledge. Mathematics becomes a model for knowledge since assertions of knowledge are credible if proof is given as logical evidence and that makes for truth and certainty. Geometry, astronomy and Aristotelian logic were the most outstanding scientific products. The practical field of knowledge most closely approaching something that can be recognized as the practically orientated science of today was medicine. In medicine, problems were solved using an ancient conceptual pattern for classification, diagnosis and cure, therapy. It was used after having made a prior study of the sequence of things up to the present stage of the problem. This linked to practical medical knowledge and it was, typically, experience based.

1.2.3 Medieval or Middle Ages: Scholastic Method

In the middle ages (500 AD - 1400), institutions in terms of knowledge and knowledge production were created. The monasteries or Universities were also created in the central Middle Ages for training masters, and subsequently doctors, lawyers, etc. Those institutions engaged in a lot of research work. This assessment is one that typically looks at them from a modern empirical, scholarly point of view. The massive technical advances that took place in the central Middle Ages resulted in the writing of practical manuals and the formulation of working knowledge for inclusion in a knowledge dissemination system. Lots of notions about academia stem from the University of the middle ages and can still be found at present. For example, Professors have the right to profess, i.e. assert knowledge; while a dissertation is still defended.

1.2.4 Industrial Revolution: Scientific Method

During the 1500s and 1600s, the concept of knowledge changed radically. By the end of 1600s, the scientific society had evolved and was expanded with new institutions during 1700s and 1800s. The scientific revolution focused on power. It is not only water and steam energy to drive the industrial revolution's factories, but also knowledge as power, thus empowering action. Knowledge can be used. The theoretical and the practical thus become interlinked and connected to the empirical, observation and experimentation resulting in effective action. A Scientific Method was formulized which included formulating hypotheses and testing hypotheses. The knowledge is tested by proven statistical means for validation. Scientific societies produced knowledge and used a peer review process to validate and disseminate new knowledge.

1.2.5 Modern Era: Learning Organization

In the modern era (post 1900), the focus is on value-adding knowledge, human and social aspects of knowledge. Knowledge has become a service. That is to say knowledge has become a part of the market-economy system rather than a prerequisite for it. Knowledge thus arises through interpretation or experience, abstraction and conceptualization. Knowledge in informatics, biotechnology and management are the great areas for the expansion of knowledge. They are closely bound up with the creation of added value in the post-industrial knowledge society. There is a transformation of a powerfully empirical and cognitively orientated concept of knowledge into a concept characterized by experience, learning and linguistic-social communality. These include:

- Continuously expanding the body of knowledge, not merely within a discipline, but within an organization's people, social networks and infrastructure, toward continuously improved performance, i.e. Learning Organization.
- How to get a team of people to collaborate effectively to enable their combined existing knowledge to be brought to bear on the problem on hand.
- How to excite innovation; the invention of new knowledge to even better, solve the problem on hand.
- A holistic view of systems rather than a mere component level view of systems. The evolution of thought from focus on components in a system, to focus on the whole (holistic) system.
- The complexity of systems due to the interaction of individual components that operate in independent and non-linear ways. The evolution of thought from a purely mechanical interpretation of activities to the recognition of a non-linear interactive model.
- How people interact in groups, especially their ability to adapt to changing situations. A shift in focus from purely individual excellence to group excellence.
- How to value different types of knowledge including intellectual capital, human, social and structural capital. Establishing metrics to evaluate success toward converting knowledge processes into improved performance.
- How to value and reward the knowledgeable, i.e. people who make use of knowledge to produce new knowledge will be superior to the manual labor of the industrial era.
- The knowledge within a community are the experts, those that

have the needed knowledge to solve or help to solve problems.

In middle ages, the focus of knowledge is on value adding and social aspects and knowledge becomes a service.

- [True](#)
- [False](#)

During industrial revolution, the concept of knowledge was changed to become interlinked and connected to hypotheses, experimentation, and observation.

- [True](#)
- [False](#)



Some Thought of Knowledge in Chinese

Confucianism has had a substantial influence on Chinese intellectual and political life. The oral and written traditions of Chinese antiquity in the codification of knowledge have been passed on down through the generations. In the Qin dynasty (213 BC), all Confucian books were burned and Confucians were persecuted as a result of the consolidation of the Qin dynasty. In the Han dynasty, Confucian works, copies of those which had been destroyed in the Qin dynasty, were taught by learned scholars in the imperial college. Unlike the practice of the Medieval Scholastic tradition, graduates of the imperial college sat for imperial civil service examinations which were based on classical literature such as "the Five Classics" and "Analects (Lun-Yü)" in order to receive their civil service appointments rather than being awarded degrees. People looking for government positions received their appointments according to the strength of their knowledge of classical literature. The imperial civil service examination was abolished only in 1905.

1.3 What is Knowledge?

As mentioned in Section 1.2, there are different concepts of knowledge during different periods of time. In addition, there are various differentiations of knowledge. One of the definitions draws from dictionary (Collins English Dictionary) is "the facts, feelings or experiences known by a person or group of people". Knowledge includes familiarity, awareness and understanding gained through experience or learning (<http://www.library.nhs.uk/knowledgemanagement/>). Although many companies are familiar with managing their operations business processes such as marketing, finance, sales, supply chain, etc., in the long term this is inadequate for success in the keenly competitive environment and fast

changing world. There is a need to innovate in order to succeed in the market. How to use their organizational knowledge to create new ideas and new knowledge becomes a critical issue. In recent years, many enterprises have started to focus on managing knowledge so as to create value for themselves.

Can knowledge be managed? It appears to be difficult to manage such an abstract thing. Knowledge is largely cognitive and highly personal, while management involves organizational processes and the ability to link them together. Many knowledge workers do not like to be managed in the traditional way and this is a barrier for KM. However, knowledge is increasingly recognized as a crucial organizational resource that gives market leverage. Today most large organizations have some forms of KM initiative. Many companies have created knowledge teams and appointed Chief Knowledge Officers (CKO). Knowledge is firmly on the strategic agenda.

Distinctions are often made between data, information, knowledge and wisdom. Knowledge is information combined with experience, context, interpretation, and reflection (Davenport et al., 1998). Knowledge is a renewable, re-usable and accumulating asset of value to firms that increases in value with employee experience and organizational life (Ginsburg and Kambil, 1999). Knowledge is intangible, boundaryless, and dynamic, and if it is not used at a specific time in a specific place, it is of no value (Nonaka and Konno, 1998). According to Fahey and Prusak (1998), knowledge is what a knower knows; there is no knowledge without someone knowing it.

1.3.1 Data, Information Vs Knowledge

Knowledge is a hierarchical network of rules about specific data or information that gives explanatory, predictive, and functional power to people. However, knowledge is neither data nor information and the differences between these terms are often a matter of degree (Davenport and Prusak, 1998). Knowledge has been defined quite differently throughout the ages. Data, information and knowledge are not interchangeable concepts.

The difference between data, information, knowledge and wisdom is illustrated in Figure 1.2.

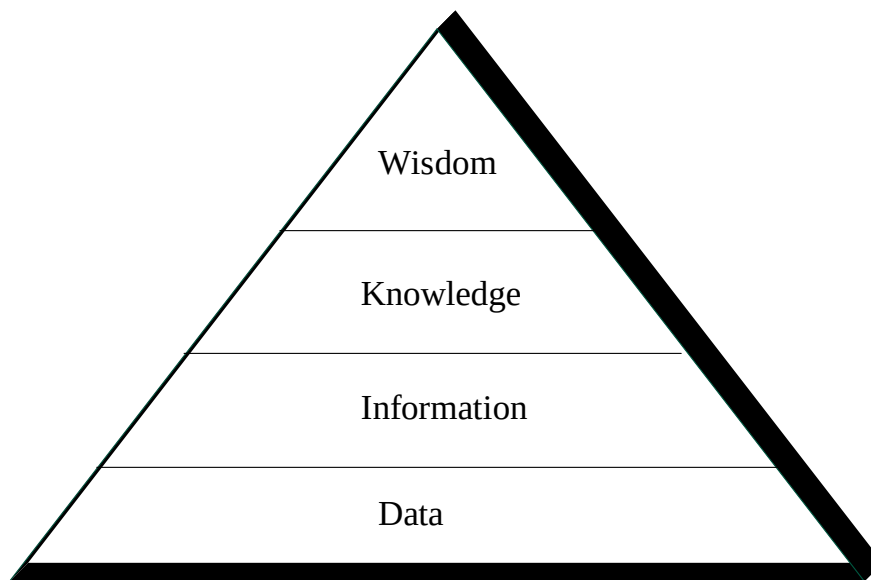


Figure 1.2 Knowledge pyramid

- **Data:** Data is a set of discrete and objective facts about events. Data is usefully described as structured records of transactions. Data by itself has little relevance or purpose. It is perceived in patterns.
Example: when a customer goes to a supermarket to buy beer, that transaction can be described by data: when he made the purchase and how many cans of beers he bought. However, the data tell us nothing about his buying behaviour. Such facts provided by data say nothing about whether the service was good or bad.
- **Information:** Information is data structured to make sense. Unlike data, information has meaning i.e. relevance and purpose. Information is organized to some purpose. Data becomes information when its creator adds meaning.
Example: the customer goes to the supermarket from Monday to Friday and the number of cans of beer he purchases (data) from Monday to Friday are 2, 3, 4, 5 and 6 respectively. The increasing trend observed from the organized purchase data becomes information describing his buying behaviour.
- **Knowledge:** Knowledge consists of patterns of information in the form of rules, which can be used to explain or predict phenomenon. It is structurally supported by information and data. Knowledge develops over time, through experience. It originates from and is applied to the minds of experts. In organizations, it often becomes embedded not only in documents or repositories but also in organizational routines, processes, practices and norms.
Example: A number of customers have the trend of buying quantities of beer that increase from Monday to Friday (information). Such an increasing trend may lead to decisions made by the supermarket manager regarding the replenishment

of stocks of beer. Besides, the supermarket manager, a keen football fan, understands that such an increasing trend is due to the World Cup football matches taking place during the next few weeks and he decided to set up a special replenishment policy during the World Cup period.

- **Wisdom:** Wisdom is proven confidence in knowledge or in decision making that is usually gained from experience and from the sharing of business focuses.

Example: The supermarket manager applied his World Cup football matches knowledge on Olympic Athens 2004 Game and had predicted the trend due to the Game and decided to set up a special replenishment policy during the Game period.

As a result, knowledge must be viewed as originating 'between the ears' of individuals. Taken literally, the need for a knower raises profound questions as to whether and how knowledge can exist outside the heads of individuals. Although knowledge can be represented in and often embedded in organizational processes, routines, and networks, and sometimes in document repositories, it cannot truly originate outside the heads of individuals. Nor is it ever complete outside of an individual.

People manage knowledge as individuals, as groups, as organizations, and as societies according to rules. These rules are categorized as procedural and declarative (or descriptive) as shown in Table 1.1.

Table 1.1 Rules for managing knowledge

	Conditions	Focus	Example
Procedural rules	Validated, hierarchical network of procedural rules	Know-how	• Cookbook • Procedural manual in an organizational environment
	Rules of how something occurs	Know-why	• Algorithms in software code • All forms of policies and plans
Declarative (descriptive) rules	Rules that give people useful, predictive, and explanatory power	Know-what	• The measurements taken by a surveyor. The surveyor's results describe some states of the world

Both types of rules work together to form human knowledge. i.e.

Procedural rules + Declarative rules = Human Knowledge

The rules of knowledge are produced and managed in knowledge processes. Knowledge processes are what individuals and collectives use

to produce, transmit, acquire, store, and use knowledge. Organizations want to become "adaptive," "agile," "nimble," and "high velocity." Good KM is a powerful way to achieve these states. Knowledge processes will be discussed in Lesson 3 and 4 in this Subject.

1.3.2 Knowledge in Action

Knowledge can and should be evaluated by the decisions or actions to which it leads. Better knowledge can lead to better decisions or actions taken. Knowledge can be used to make wiser decisions about strategy, competitors, customers, product and service life cycle. Since knowledge usually resides in people's heads, it is worth reflecting on some of its key components such as experience, truth, judgment, rules of thumb, values, beliefs and assumptions.

- (i) **Experience:** Experience refers to what has been done and what has happened in the past. Knowledge develops over time through learning. Experience provides a historical perspective from which to view and understand new situations and events. For example: when companies hire experts, they are buying experience-based insights.
- (ii) **Judgment:** Unlike data and information, knowledge contains judgment. Knowledge is used to judge new situations and information in light of what is already known and is used also to re-examine and refine itself in response to new situations and new information. Knowledge is growing and changing as it interacts with the environment and information provided. When knowledge stops evolving, it turns into opinion. Judgment allows knowledge to rise above and beyond an opinion when it re-examines and refines itself when it is applied and acted on in order to handle situations. For example, a stock broker makes a sell and buy judgment based on the data and information provided. This knowledge is of no value if the stock broker can't act on it or make a decision based on the data and information provided. In other words, unactionable information is not knowledge
- (iii) **Ground truth:** Experience changes ideas about what should happen into knowledge of what does happen. Ground truth means knowing what really works and what doesn't. Knowledge contains ground truth to describe the rich truths of real situations experienced rather than from theory or generalizations. In other words, people share the detail and meaning of real experiences because knowledge contained in the messy reality of daily work is generally more valuable than theories.
- (iv) **Values and Beliefs:** Values and beliefs are integral components of knowledge, determining in large part what the knower sees, absorbs, and concludes from his observations. People's values and beliefs have a powerful impact on organizational knowledge. Organizations are made up of people whose values and beliefs affect their thoughts and actions. For example, profits and market

dominance are an integral part of Microsoft. These beliefs and values are inherent in the very character of Microsoft.

Discuss your definition of knowledge in Bulletin Board.



What is Knowledge?

- A. [Tacit and explicit mental actionable understandings and capabilities in people and structural intellectual capital \(IC\) in organizations.](#)
- B. [Data and information that describes rules and relationships.](#)
- C. [Documented instructions and information for how to do things and conduct work.](#)
- D. [Information in corporate knowledge bases.](#)



1.4 Types of Knowledge

Knowledge can take many forms, states, and modes, and knowledge can have many attributes and can exist in explicit and in tacit forms.

1.4.1 Explicit Knowledge

Explicit knowledge is visible knowledge. It is objective and rational knowledge that can be expressed in words, sentences, numbers or formulas (context free). Explicit knowledge includes patents, procedures, best practices, and lesson learned. Explicit knowledge is relatively easy to capture and store in databases and documents. It includes theoretical approaches, problem solving, manuals and databases and the vast sea of written and verbal information recorded in video, audio, and written formats.

There are billions of lines of software code with scientific and corporate procedural knowledge taken from countless interviews and validations. IT technology can help us to extract knowledge from codified sources. Unlike tacit knowledge, once codified, it is much more easily captured, stored and transferred to others. This also can be shared with a high degree of accuracy.

1.4.2 Structured and Unstructured Knowledge

Explicit knowledge can be categorized as either structured or unstructured. The individual data elements of structured knowledge are organized in a particular way or schema for future retrieval.

There is an abundance of unstructured knowledge such as text in our

daily life and organizations. Art Group's recent research into the subject of unstructured knowledge has revealed that 80-98% of all data in all computers is unstructured knowledge.

1.4.3 Tacit Knowledge

Tacit (implicit) knowledge is unseen and resides primarily in the heads of knowing individuals. It includes mental models, and both "know-how" and "know-why". It is obscure and difficult to access. Tacit knowledge is subjective, and experience based knowledge that can not be expressed in words, sentences, numbers or formulas, often because it is context specific. It is the unspoken, non-codified sum of all the cognitive skills such as beliefs and experience and know-how of individuals. When a firm look for ways to extract knowledge from an employee or consultant before he leaves the company or changes positions, the firm is trying to recover knowledge that would take time and money to replace.

Tacit knowledge is also a primary source of new knowledge in the knowledge production process. By careful observation, questioning, and validation, the unrecorded know-how of an individual can be codified and turned into new knowledge. Tacit knowledge is considered more valuable because it provides a context for people, places, ideas and experiences. Tacit knowledge generally requires extensive personal contact and trust if it is to be shared effectively.



A market survey report is a piece of explicit knowledge.

- [YES](#)
- [NO](#)



The cooking skill of a chief can be classified as explicit knowledge.

- [YES](#)
- [NO](#)

1.5 Modes of Knowledge Conversion

The Nonaka and Takeuchi knowledge spiral/conversion model (or so call SEIC cycle of knowledge conversion) is basically a two-dimensional matrix depicting four possible scenarios of tacit and explicit knowledge interaction or "conversion". As shown in Figure 1.3, Nonaka models of knowledge transfer as a spiral process in which existing knowledge can be in either form i.e. tacit or explicit. Each type of knowledge can be converted. The objective of knowledge transfer can be to convey either tacit or explicit knowledge and each mode of transfer operates differently.

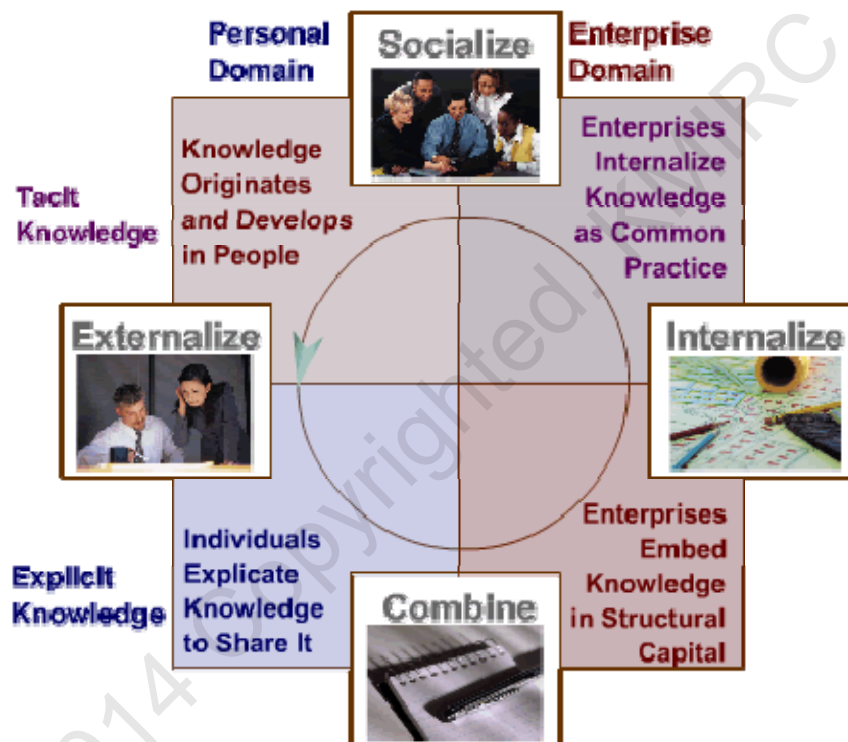


Figure 1.3 Nonaka's Four Modes of Knowledge Conversion

When viewed as a continuous learning process, the model becomes an anti-clockwise spiral. Organizational learning depends on initiating and sustaining the learning spiral. The model is a spiral but not a cycle. This is because as one "Learns" around the spiral, understanding moves to deeper and deeper levels:

1.5.1 Externalization

The process for making tacit knowledge explicit is externalization. One case is the articulation of one's own tacit knowledge and the other is translating the tacit knowledge of others into a readily understandable form. During dialogue, people can share beliefs and learn how to articulate their thinking better, through instantaneous feedback and the simultaneous exchange of ideas. Externalization is a process among individuals within a group. E.g. writing a case study and reports.

Example:

General Electric (GE) records all customer complaints and enquiries so they can be used by members of new product development teams to re-experience what the customer service staff experienced.

1.5.2 Internalization

Internalization is a process of understanding and absorbing explicit knowledge thus turning it into tacit knowledge held by the individual. Knowledge in the tacit form is actionable by the owner via actual doing or through simulations. The internalization process transfers organization and group explicit knowledge to the individual. E.g. learning a skill through training and practice.

Example:

Mazda combined deduction and induction to create the RX-7 concept. The concept was deduced from the car maker's corporate slogan. At the same time, the new concept was induced from concept trips which were driving experiences by development team members in the US as well as from opinions gathered from customers and car experts.

1.5.3 Socialization

The process that transfers tacit knowledge in one person to tacit knowledge in another person is socialization. It is primarily a process between individuals. It involves capturing knowledge by direct interaction and sharing experience with individuals outside and inside an organization. This results in acquired skills and common mental models. E.g. apprenticeship, story telling and experience sharing from lessons learned.

Examples:

- Reading and listening to a success story makes some members of the organization feel the realism and essence of the story. The experience that took place in the past may change into a tacit mental model.

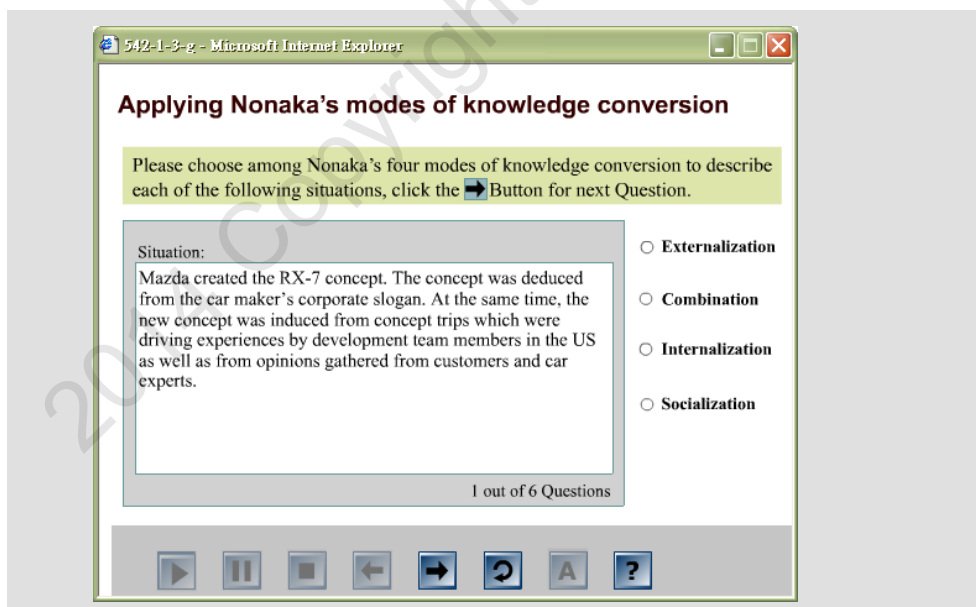
- Honda set up brainstorming camps – informal meetings for detailed discussions to solve difficult problems in development projects. The meetings included more than just project members. Brainstorming techniques were used. Meetings included much that might be categorized as trust and morale building.
- Matsushita Electric Industrial Company was unsuccessful at developing an automated bread-making machine. They knew the best bread came from the Osaka International Hotel. As a result, a few team members apprenticed themselves to the hotel's head baker.

1.5.4 Combination

Combination is a process of converting explicit knowledge into a more usable means of explicit knowledge. Since explicit knowledge can be conveyed in documents, emails, databases, it can be made more usable by collecting, disseminating and editing/processing it. This is the area where information technology is most helpful. Combination allows knowledge transfer among groups across organizations. E.g. document management system.

Example:

Kraft General Foods utilizes data from its Point-of-Sales (POS) system not only to find out what does and does not sell but also to create new sales systems and methods.



What do you think the Nonaka and Takeuchi knowledge spiral/conversion model is?

- [An illustration of how tacit personal knowledge through a series of transformations becomes useful structural](#)



[intellectual capital.](#)

- [A model for how knowledge is created.](#)
- [A model for how creativity and innovation can be fostered.](#)
- [An illustration of how organizational knowledge is applied to achieve better performance.](#)

1.6 Meta-level of Knowledge

Knowledge can be viewed in levels. Level 0 represents the world people interact with. It does not contain knowledge itself. However, knowledge can be acquired about the world by interacting with the world and through evolution and/or learning. Level 1 knowledge is the knowledge that one uses to interact with the world. People interact, learn and apply knowledge and the cycle continues when they are doing their jobs or tasks.

People can reflect on ways to improve the way they learn. Better decisions can be made based on that reflection of learning. A level 2 process is related to how people reflect on how they produce, acquire and transfer knowledge. This is a knowledge management process. In order to make level 2 processes better, a level 3 reflection process is required in which people manage knowledge about knowledge management, i.e. the role of a Chief Knowledge Officer or a KM Consultant, etc.

Meta-cognition is awareness of one's own thinking process. Meta-data is data about data. Meta-logic is the logic about logic. The concept of Meta-data will be discussed in *Methods and Tools for Knowledge Management Systems* (ISE543) in detail. As regards knowledge:

Level 0 is the world.

Level 1 is knowledge about the world

Level 2 is knowledge about knowledge about the world

Level 3 is knowledge about level 2

⋮

Level N depicts the fact that there is no limit to the levels. Any level can have a meta-level.

Two application examples are shown in Figure 1.4 and Figure 1.5 respectively.



Figure 1.4 Meta-levels of knowledge for a manufacturing process

This is a manufacturing process example.

Knowledge at level 0 is represented by the products produced and sold to customers. This includes the workers, machines, parts, raw materials, etc.

Knowledge at level 1 is where workers are applying their knowledge to manufacture those products and services. The manufacturing process is the firm's knowledge for manufacturing the products.

Staff move into knowledge at level 2 when workers or managers reflect on how to improve that manufacturing process.

Should they reflect on how to improve the process of improving processes, they move into knowledge at level 3 and so on.

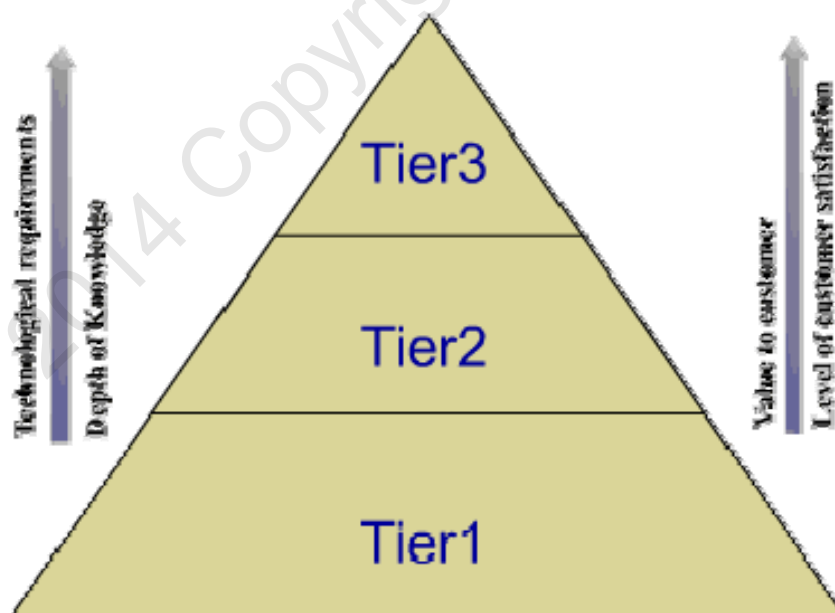


Figure 1.5 Knowledge Pyramid for CSM

This is a customer service management (CSM) example. A knowledge pyramid for customer services is shown in Figure 1.5. Basically, there are three tiers of knowledge domains for customer services which are Tier one, Tier two and Tier three, respectively (Cheung et al., 2003). Tier one relates to general data and information services while Tier two focuses on advisory oriented services. Tier three is concerned with knowledge and the expert level of customer service.

Tier one domain of customer services (Level 0), relates to the general data and information services provided to the customers. Such data and information are explicit knowledge for the customers which support their decision making.

Example: Tier one domain is most commonly found in most of the voice-enabled and automated call-centres used in banks, insurance and telecommunications companies. They usually provide data, facts and information for the customer's references. The customers are presumed to possess prior knowledge related to their problems and the services provided are only for helping them to make decisions.

Tier two domain of customer services (Level 1) relates to the advisory services in which the customers are provided with solutions, suggestions and recommendations to solve the problems.

Example: Examples can be found in the automatic help desk services provided to computer users. The solutions and recommendations are usually derived or deduced from data, information and facts in the Tier one domain of knowledge (Level 0), i.e., a meta-level of knowledge of Tier one. In other words, the level 1 knowledge of customer service is a reflection of the knowledge in level 0.

The Tier three domain of customer services (Level 2), relates to specific knowledge and expert services. It is commonly found in consultancy businesses which provide consultancy services to their customers who are also practitioners in the field and possess prior knowledge. It is interesting to note that the knowledge for Tier three domain of customer services involves a meta-level of knowledge of Tier two (Level 1) and Tier one (Level 0) of knowledge domain, i.e. knowledge used to manage the knowledge of customer service staff.

Example: For example, a consultancy company manages and trains up the expertise of its consultants so they are competent to provide data, information and advisory services (Level 0 and Level 1 of knowledge) to their customers. This is a knowledge management process.

1.7 Intellectual Capital

Companies with high levels of market valuation are often companies with high levels of intangible assets which is referred as intellectual capital (IC). IC might be any asset that cannot be measured but is used by a company to its advantage. A company's expertise and their

competencies, goodwill, patents, corporate brands, distribution channels, contracts, collaborators, leadership, customer base, trade marks, company culture, and reputation are some of examples of IC.

As business shift from an asset-centric environment to a knowledge-centric environment, traditional value measures become inadequate. In a knowledge-based economy, the security is the value of its intangible assets and their perceived future value. Most of high potential growth companies, they are relaying more on knowledge and people. A lot of companies with "real" assets such as building, manufacturing facilities, equipment and offices possesses value far lesser than their market valuation.

As shown in Figure1.6, the ratio between market capitalization and physical asset for different companies such as Johnson Electric (JE), Microsoft and IBM are much higher than 1. This implies that the knowledge in those companies can be converted into value. It is shown that the successful implementation of the Centre of Excellence (COE) system in JE helps the knowledge-centric environment to leverage its intangible asset and market capitalization.

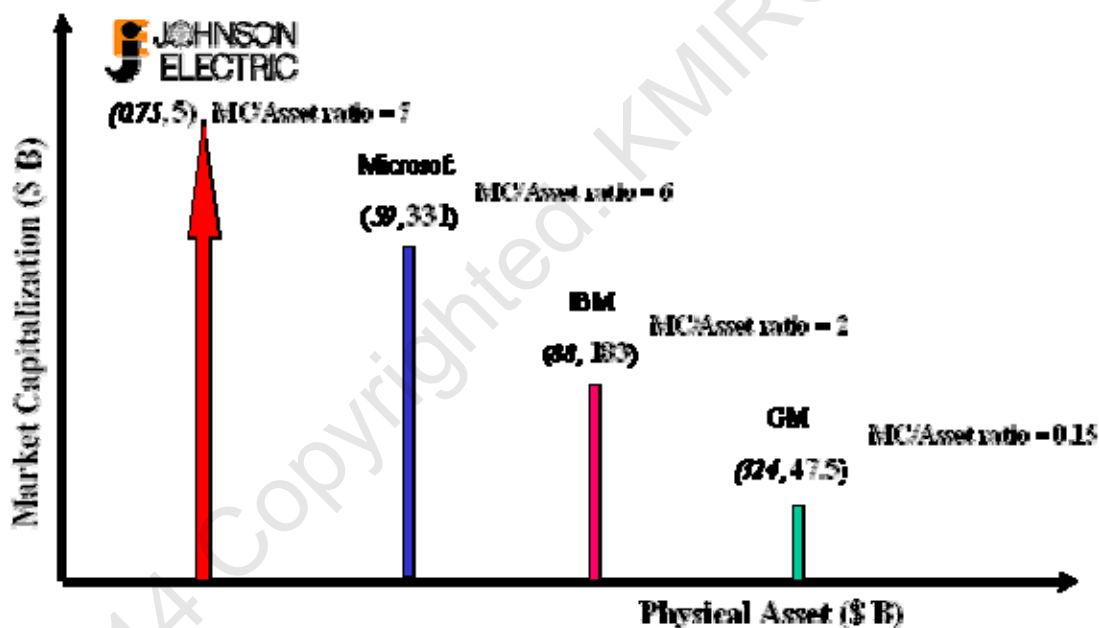


Figure 1.6 Market Capitalization Vs Physical Asset (Ko et al. 2004)

Another example is for Netscape Corporation which went to public a few years back. The market valued this US\$17 million company at US\$3 billion at the end of the very first day of trading. The market did not value the company on the basis of its buildings and computers but on the basis of its knowledge assets: its invention of the Web browser, its patented technology, etc.

More details about measuring and managing IC will be discussed in Subject *Managing and Measuring Intellectual Capital* (ISE548).

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